

WEALTHWISE

Smart Savings & Tax Planning Platform for India

Project Documentation — Version 5.2

Field	Value
Financial Year	FY 2026-27
Platform	Web App + PWA (Android-installable)
Tech Stack	React 18, FastAPI, Firebase, Claude AI (Anthropic)
Current Version	5.2
Theme	Kuber / Laxmi — Saffron Gold & Temple Emerald
Status	Active Development

1. Executive Summary

WealthWise is an India-specific, AI-powered personal finance web application that helps individuals make smarter savings and investment decisions while minimising their income tax liability under FY 2026-27 rules.

Unlike generic financial advisory tools, WealthWise is built from the ground up for the Indian market — incorporating actual IT Act sections, 14 Indian investment instruments, HUF entity support, agricultural income handling, multi-source income aggregation, and the latest Budget 2025 changes including the Section 87A rebate up to ₹12 lakh taxable income.

The platform combines a deterministic multi-income tax engine, a scoring-based risk profiler, a goal-aware portfolio allocator, an IT deductions reference with 35+ sections, a monthly expense tracker with 8 Indian bank PDF parsers, a financial goals SIP planner, and a What-If scenario planner — all integrated with Anthropic Claude AI for plain-English guidance.

2. Problem Statement

2.1 The Gap in Existing Solutions

A significant portion of Indian salaried and self-employed individuals lack access to personalised, affordable financial planning. Seven comparable open-source projects were systematically analysed:

Repository	Core Approach	Critical Gaps
Mugeshwaran-k / Financial-Ai-Advisor-Bot	LLM chatbot (Clerk auth)	No tax logic. API key exposed in frontend.
Annarhysa / BudgetBud	Expense tracker with limits	No investment advice. No tax engine.
Aaditya-5281 / Finance-Advisor	Generic LLM Q&A	No regime comparison. No calculations.
marinaredamekhael / FinancialAdvisor	Rule-based budget calc	USD-centric. No India context.
hadil1999-creator / RAG_Hack_team	RAG over finance PDFs	Over-engineered. No live data. No portfolios.
imRahul05 / Fin_service	India-focused LLM wrapper	No actual tax slab code. Generic prompting only.
EnigmAI / Financely	ML sentiment + KNN	Non-Indian data. API key insecurity.

2.2 Specific Flaws Resolved by WealthWise

- No existing project implements FY 2026-27 Indian income tax slabs, regime comparison, or Section 87A rebate logic in deterministic code — all rely on vague LLM prompting.
- None cover India-specific instruments: PPF, ELSS, NPS, SGB, NSC, SCSS, or HUF entity-specific tax treatment.
- Five of seven projects expose API keys in frontend code. WealthWise uses a backend proxy — the Anthropic key never leaves the Railway server.

- No project handles multi-source income (salary + rental + FD interest + LTCG + agricultural + crypto) in a single tax computation.
- None implement agricultural income via the partial integration method or loan deductions as deduction sources.
- No project provides expenditure-based tax deductions — WealthWise auto-extracts 80D and 80C hints from the expense tracker.
- None provide a comprehensive IT Act deductions reference covering 35+ sections with limits, regime applicability, and pro tips.
- No project provides recurring transaction auto-posting, bank PDF statement import, or per-category budget alerts.
- RAG pipelines add unnecessary complexity. Claude API with a structured FY 2026-27 system prompt achieves superior accuracy with zero infrastructure overhead.

3. Objectives

1. Build a fully India-specific multi-income tax engine for FY 2026-27 covering 16 income source types.
2. Support HUF (Hindu Undivided Family) as an entity type alongside Individual, with correct tax treatment.
3. Provide personalised portfolio allocation across 14 Indian and global asset classes with goal-aware adjustment.
4. Show both pre-tax and post-tax CAGR per asset per regime, and dual corpus projections (new vs old regime).
5. Create an exhaustive IT Act deductions reference covering 35+ sections with an AI advisor backed by Claude Sonnet.
6. Implement a monthly income and expense tracker with expenditure-based tax deduction extraction (80D, 80C hints).
7. Build a recurring transaction engine with auto-posting, CSV import from 8 major Indian banks, and per-category budget alerts.
8. Deliver a financial goals module with automatic SIP calculation and instrument recommendation, integrated into the planner.
9. Build a What-If scenario planner with side-by-side dual projections and milestone snapshots.
10. Ensure production-grade API key security: backend proxy, rate limiting, CORS restriction, spend cap.
11. Deploy as a PWA installable on Android.
12. Build a personalised ITR Filing Guide (FY 2026-27 / AY 2027-28) that auto-determines the correct ITR form (ITR-1/2/3/4) from the user's income profile, generates a dynamic document checklist, maps income sources to exact portal schedules, provides step-by-step incometax.gov.in portal walkthrough, calculates Sec 234F late filing penalties.

4. System Architecture

4.1 High-Level Architecture

The system follows a three-tier architecture with clear separation between the client, backend, and external services:

Layer	Technology & Role
React Frontend (Vercel CDN)	UI rendering, browser-side tax calculations, PWA
Firebase Auth + Firestore	User authentication (Google/Email), per-user data storage
FastAPI Backend (Railway)	Claude AI proxy, rate limiting, CORS, multi-income API
Anthropic Claude Sonnet	Portfolio explanations, tax Q&A, savings coaching

4.2 Security Architecture

- ANTHROPIC_API_KEY stored exclusively in Railway environment variables. Never in frontend code or GitHub repository.
- All AI calls go through /ai/* FastAPI routes. Browser never contacts Anthropic directly.
- Rate limiting via slowapi: 10/min for portfolio explanations, 15/min for tax advisor, 60/min for /analyze.
- CORS locked to production Vercel domain via ALLOWED_ORIGINS environment variable.
- Firebase Security Rules: users can only read/write their own Firestore documents (matched by Firebase UID).
- Monthly spend cap set at console.anthropic.com to limit maximum financial exposure.

4.3 Firestore Data Architecture (ER Diagram)

WealthWise uses Firestore with a hierarchical sub-collection structure under each user document:

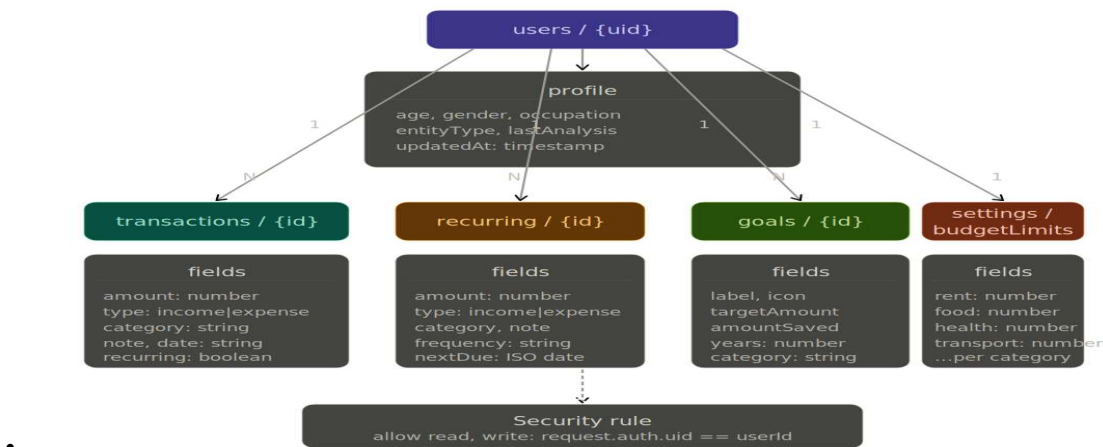


Figure 1: WealthWise Firestore Entity-Relationship Diagram

Each user document (users/{uid}) contains a profile sub-document and four sub-collections:

- transactions/{id} — individual income/expense records with category, amount, date, note, recurring flag
- recurring/{id} — recurring transaction templates with frequency and nextDue date
- goals/{id} — financial goals with target, timeline, saved amount, and SIP calculation
- settings/budgetLimits — per-category monthly budget limits (single document)

5. Application Screenshots

5.1 Portfolio Results — Pre/Post-Tax Corpus

Results show risk profile, total tax under best regime, and 10-year corpus under both regimes. Each asset displays pre-tax CAGR → post-tax CAGR (new) / post-tax CAGR (old) with the applicable tax rule. The projection chart shows three lines: pre-tax, new regime, old regime.



Figure 3: Portfolio Results — Aggressive profile, 10-year corpus with tax drag

5.2 Income Sources — Multi-Source Input

The income form accepts 16 income source types across four groups: Ordinary Income (slab rate), Capital Gains (special rates), Special Income (exempt or 30% flat), and Loans & Deductions (old regime only).

Ordinary Income

Slab Rate

Salary / CTC

Enter your TOTAL annual CTC — includes ALL components (basic, HRA, DA, bonus, allowances). Do NOT subtract HRA here. HRA exemption is computed separately via the HRA sub-section below.

Business Profit

Enter NET profit after all business expenses. OR toggle 44AD Presumptive below — enter gross turnover and 8% (or 6% digital) is auto-deemed as Income (no bookkeeping needed for turnover ≤ ₹2Cr).

Futures & Options (F&O)

Enter NET P&L — profit minus loss. F&O is treated as business Income under ITR-3. Also enter F&O gross turnover in the expanded panel for the Section 44AB audit threshold check.

Freelance / Consulting

Enter NET receipts after expenses. Professionals (doctors, CAs, lawyers, engineers) with gross receipts ≤ ₹75L can opt for 44ADA Presumptive — 50% is auto-deemed income, no detailed books needed.

Rental Income

Enter GROSS annual rent received. 30% standard deduction is auto-applied (Sec 24). Municipal tax paid is also deductible — enter net of that.

FD / RD / Post Office Interest

Fully taxable at slab. TDS @10% deducted by bank if interest >₹40K/yr (₹50K for seniors).

Savings Account Interest

80TTA: first ₹10K exempt (under 60). 80TTB: first ₹50K exempt (60+). Auto-applied based on your age. Enter full interest received.

Dividend Income

Taxable at your slab rate. TDS @10% above ₹5K per company.

Other Income

Pension, gifts above ₹50K, winnings, lottery, etc. Taxed at your marginal slab rate.

Capital Gains

Special Rates

LTCG — Equity / Equity MF

10% tax on gains above ₹1.25L exemption (₹25-27). Enter GROSS LTCG before the exemption. Equity shares or equity-oriented MF held >1 year.

STCG — Equity / Equity MF

15% flat on the full gain. No exemption. Equity or equity MF held <1 year.

LTCG — Debt MF / Bonds (Sec 112)

Debt MF or bonds purchased BEFORE April 2023, held >3 years. Sec 112: 20% with indexation. Enter the gain AFTER applying your indexation adjustment (use CII table from [Incometax.gov.in](#)).

LTCG — Property/Gold (Pre Jul 23, 2024)

20% WITH Indexation. Property or gold purchased before July 23, 2024. Sec 54/54F/54EC exemptions may apply — see IT Guide.

LTCG — Property/Gold (Post Jul 23, 2024)

12.5% WITHOUT Indexation. Budget 2024 rule for property/gold acquired after July 23, 2024.

Special Income

Exempt / 30%

Agricultural Income

Exempt from central income tax. Still declare — it is used for rate computation (partial integration) if total income exceeds basic exemption.

B Crypto / VDA Income

30% flat tax + 1% TDS per transaction. No deductions allowed, no loss set-off against any other head.

Figure 2: Income Sources Form — HUF selector, 13 income types, live tax preview

5.3 Expense Tracker — Dashboard

The tracker dashboard shows monthly KPIs, spending breakdown pie, 6-month overview bar chart, filter controls, and the AI Savings Coach. Budget alerts appear at the top when any category approaches its limit.



Figure 4: Expense Tracker — Dashboard with KPIs, pie chart, filter, and AI coach

5.4 ITR Filing Guide

The ITR Filing Guide provides personalised guidance for FY 2026-27 (AY 2027-28). It auto-determines the correct ITR form (ITR-1 for simple salary, ITR-2 for capital gains/crypto, ITR-3 for business/F&O, ITR-4 for presumptive income). Six tabs: ITR Form, Schedules, Documents, Filing Steps, Deadlines, and AI Assistant.

ITR Filing Guide

Personalized guidance for FY 2026-27 (AY 2027-28) — which form to file, documents needed, and step-by-step portal walkthrough. **Fill the Planner section first for personalized guidance.**

↑ **To personalize this guide:** Go to Planner → add your income sources → run analysis. This section will then auto-determine your ITR form, show only your relevant documents, and map your exact schedules on the portal. The guide below shows general information applicable to all.

ITR Form | Schedules | Documents | Filing Steps | Deadlines | AI Assistant

ITR-1 (Sahaj) Simple

Online on portal — no offline utility needed

Who should file

- Resident individual (not NRI)
- Salary or pension income
- One house property
- Other sources: FD interest, savings interest, dividends
- Agricultural income ≤ ₹5,000
- Total income ≤ ₹50 lakhs

Not applicable for

- Any capital gains (LTCG / STCG)
- Business or profession income
- F&O trading
- Crypto / VDA income
- Foreign income or foreign assets
- Company directors
- Total income above ₹50L

Portal path: incometax.gov.in → e-File → File ITR → ITR-1 → Online

All ITR Forms — Quick Reference

ITR-1	ITR-1 (Sahaj) Resident individual (not NRI)	Simple
ITR-2	ITR-2 Individuals/HUF NOT having business income	Moderate
ITR-3	ITR-3 Individuals/HUF having business or profession income	Complex
ITR-4	ITR-4 (Sugam) Presumptive business income — Sec. 44AD (turnover ≤ ₹2Cr)	Simple

Common Filing Mistakes to Avoid

- ✗ Not reconciling AIS/Form 26AS before filing — biggest cause of IT notices
- ✗ Choosing wrong Assessment Year — AY 2027-28 for FY 2026-27 income
- ✗ Using ITR-1 despite having capital gains or crypto income
- ✗ Reporting F&O P&L under capital gains instead of business income (Schedule BP)
- ✗ Forgetting crypto TDS (1% per sell) in Form 26AS — mismatches trigger scrutiny
- ✗ Not declaring employer NPS (BOCCD(2)) — available even in new regime
- ✗ Mixing up old and new regime at last minute — locked after submission
- ✗ Not e-verifying within 30 days of filing — return becomes invalid
- ✗ Missing grandfathering computation for equity bought before Jan 31, 2018
- ✗ Not claiming Section 54/54F/54EC exemption when reinvesting property LTCG

Figure 5: ITR Filing Guide — Form determination, document checklist, and 8-step portal walkthrough

5.5 Import CSV / PDF Statement

The Import tab supports CSV/Excel export from SBI (merged-cell format), HDFC, ICICI, and generic CSVs, plus direct PDF parsing for 8 major Indian banks. Transactions are auto-categorised from merchant names.

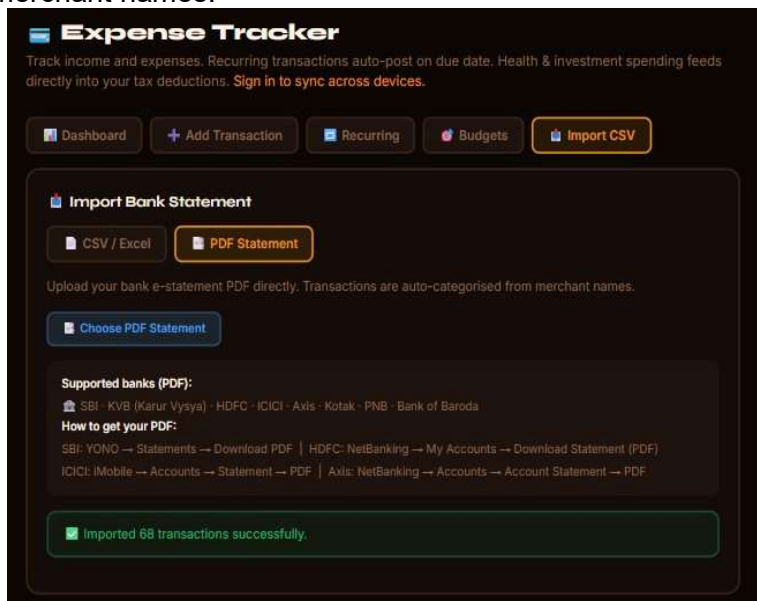


Figure 5: Import — PDF Statement mode showing 68 transactions imported from SBI PDF

6. Feature Specification

6.1 Multi-Income Tax Engine (FY 2026-27)

The tax engine handles 16 distinct income source types, computes tax under both regimes separately, and returns per-regime marginal slab rates for accurate post-tax corpus projections.

Income Source	Tax Treatment	Notes
Salary / CTC	Slab rate	Std deduction ₹75K (new) / ₹50K (old) auto-applied
Business Profit	Slab rate	Net profit after all business expenses
F&O / Futures & Options	Slab rate	Treated as business income under ITR-3. Enter net P&L. Gross turnover tracked for Sec 44AB audit threshold (₹1Cr / ₹10Cr digital).
Freelance / Consulting	Slab rate	Net receipts after deductible expenses
Rental Income	Slab rate	30% std deduction under Sec 24 auto-applied
FD / RD / Post Office Interest	Slab rate	TDS @10% above ₹40K/yr (₹50K seniors)
Savings Account Interest	Slab rate	80TTA ₹10K exempt (<60); 80TTB ₹50K exempt (60+)

Dividend Income	Slab rate	TDS @10% if >₹5K per company
LTCG - Equity / MF	10% above ₹1.25L	FY26-27 exemption raised from ₹1L to ₹1.25L
LTCG — Debt MF / Bonds (Sec 112)	20% with indexation OR 10% without	For debt MF purchased before April 1, 2023, held >3 years. User chooses lower tax option. Bonds at 10% flat.
STCG - Equity / MF	15% flat	No exemption. Holdings <1 year.
LTCG – Property (Pre Jul 23, 2024)	20% with indexation	Held >2 years. Properties purchased/sold before July 23, 2024. Sec 54/54F/54EC exemption may apply.
LTCG – Property/Gold (Post Jul 23, 2024)	12.5% without indexation	Budget 2024 removed indexation for properties purchased after July 23, 2024.
Agricultural Income	Exempt + rate computation	Partial integration method: exempt but shifts slab
Crypto / VDA	30% flat + 1% TDS	No deductions, no loss set-off, no exemption.
Other Income	Slab rate	Pensions, gifts >50K, winnings, etc.

Tax Feature	Detail
New Regime Slabs	0% ₹0-4L 5% ₹4-8L 10% ₹8-12L 15% ₹12-16L 20% ₹16-20L 25% ₹20-24L 30% above ₹24L
Sec 87A Rebate (New)	Zero tax if net taxable ≤ ₹12L (Budget 2025). Gross CTC ≤ ₹12.75L = zero tax.
Standard Deduction	₹75K new regime / ₹50K old regime. Auto-applied.
HUF Treatment	Same slabs as individual. No senior citizen benefit. Interest exemption ₹50K.
Per-Regime Slab Rates	NEW: New and old regime each get their own marginal slab rate, producing two different post-tax corpora (fixes identical corpus glitch).
HRA Exemption (Sec 10(13A))	Computed via 3-limit formula: min(HRA received, 50%/40% of basic, rent – 10% basic). Metro/non-metro toggle. Sub-fields in Income Form for HRA component, annual rent paid, city type.
Sec 44AD Presumptive (Business)	Toggle in Income Form. Deemed income = 8% of turnover (6% if 95%+ digital). Turnover ≤ ₹2Cr. No bookkeeping required.
Sec 44ADA Presumptive (Professionals)	Toggle in Income Form. Deemed income = 50% of gross receipts. Eligible: doctors, CAs, lawyers, engineers. Receipts ≤ ₹75L.
Sec 44AB Audit Check (F&O)	Live warning in Income Form when F&O gross turnover > ₹1Cr (or ₹10Cr digital). Deadline: Sep 30.
Sec 10(14) Children Allowances	Education ₹100/mo + Hostel ₹300/mo per child, max 2 children = ₹9,600/yr exempt. Sub-field in Income Form.
Sec 89(1) Salary Arrears	Arrear amount + prior FY sub-fields in Income Form. Warning to file Form 10E before ITR

6.2 Asset Classes (14 Total) with Tax Rules

Every asset carries its tax rule. Post-tax CAGR is calculated per regime using that regime's marginal slab rate.

Asset	Pre-Tax CAGR	Tax Rule	Post-Tax @ 30%	Lock-in
PPF	7.1%	Tax-Free (EEE)	7.1%	15yr
Savings Account	3.5%	Slab Rate	2.45%	None
Fixed Deposit	7.2%	Slab Rate	5.04%	1-5yr
NPS	11.0%	Partly Tax-Free	~9.7%	Till 60
Debt Mutual Fund	7.5%	Slab Rate	5.25%	None
Digital Gold / ETF	11.0%	LTCG 12.5%	9.625%	None
Sovereign Gold Bond (SGB)	11.0%	Tax-Free (maturity)	11.0%	8yr
ELSS (Tax Saver MF)	14.0%	LTCG 10%	12.6%	3yr
Index MF (Nifty 50)	13.0%	LTCG 10%	11.7%	None
Large Cap MF	14.5%	LTCG 10%	13.05%	None
Mid / Small Cap MF	17.0%	LTCG 10%	15.3%	None
Direct Stocks	15.0%	LTCG 10%	13.5%	None
Real Estate	9.5%	LTCG 20%	7.6%	5yr+
Bitcoin / Crypto	35.0%	30% Flat	24.5%	None

6.3 Expense Tracker — v5 Features

- Recurring Transactions: toggle on add, auto-posts overdue recurring templates on app load.
- Pagination + Filtering: search by note, filter by type/category/date range, paginated in batches of 20.
- CSV Export: downloads filtered transactions as .csv with date, type, category, amount, note.
- Budget Alerts: per-category monthly limits with 90% warning and 100% exceeded alert banners.
- CSV Import: auto-detects SBI Excel export (merged-cell fixed-column format), HDFC, ICICI, and generic CSVs.
- PDF Import: 8 Indian bank parsers (SBI, KVB, HDFC, ICICI, Axis, Kotak, PNB, Bank of Baroda). Text extracted via PDF.js, grouped by Y-coordinate to reconstruct rows.
- Edit Transactions: every transaction row has an edit button (✏️) opening an EditModal to change type, amount, category, date, note.
- Empty States: helpful illustrated empty states for no transactions, no filter results, no recurring templates, no budgets.

- Family Category: added for household/dependent expenses alongside 10 other expense categories.
- Auto-Categorisation: `autoCategory()` function categorises from merchant name. KIIT Hospitality → food; personal UPI transfers → misc/personal.

6.4 Goals + SIP Calculator

- 8 pre-built templates: House, Car, Education, Wedding, Travel, Retirement, Emergency Fund, Business.
- SIP formula: $SIP = (Target - Saved) \times r / ((1+r)^n - 1)$ where r = monthly rate, n = months.
- Instrument recommendation: maps goal timeline to optimal asset (FD for <1yr, Index MF for 7yr+).
- Goals loaded into Planner: portfolio automatically adjusted based on goal timelines.

6.5 IT Deductions Reference (35+ Sections)

Each section includes exact limit, regime applicability, eligible instruments, conditions, pro tip, and an "Ask AI about this section" button that calls the backend tax advisor with user context.

6.6 ITR Filing Guide (FY 2026-27)

The ITR Filing Guide (ITRFiling.jsx) is a full-feature personalized guide with six tabs:

- **ITR Form Tab:** Auto-determines ITR-1/2/3/4 from the user's income profile passed from the Planner. Reasons shown (e.g., "F&O income → ITR-3 mandatory"). Expandable comparison table for all four forms.
- **Schedules Tab:** Maps each active income source to its exact portal schedule (e.g., crypto → Schedule VDA, property LTCG → Schedule CG Part B). Shown only for the user's income types.
- **Documents Tab:** Dynamic checklist generated per income source — only shows documents relevant to the user. Groups: Always Required, Salary, Rental, Interest, Capital Gains, Crypto, Agricultural, Business.
- **Filing Steps Tab:** 8-step portal walkthrough (Pre-verification → Login → Start Filing → Profile → Schedules → Deductions → Tax Payment → E-Verify). Each step expandable with exact portal paths.
- **Deadlines Tab:** Filing deadlines (Jul 31, Sep 30, Dec 31), Sec 234F late penalty calculator, advance tax schedule, and other key dates.
- **AI Assistant Tab:** Claude-backed filing assistant with 8 quick-question chips. Context-aware when Planner data is present (income sources, recommended form, best regime, total tax).

7. Technical Implementation

7.1 AI Integration Design

Approach	Build Time	Accuracy	Cost	Chosen?
Train custom model	3-6 months	Low	GPU servers	No
RAG pipeline (vector DB)	2-3 weeks	Good	Vector DB + embeddings	No
Claude API + system prompt	1 day	Excellent — native FY26-27 knowledge	Pay-per-call	YES

7.2 Frontend Architecture

File	Purpose
App.jsx	Main app: 3-step planner, 5-page routing, bottom nav, savings toggle, goal loading
IncomeForm.jsx	Multi-source income entry. SafeInput, SrcGroup, and OrdinaryCard hoisted to module scope (cursor/scroll fix). v5.1: OrdinaryCard handles salary sub-fields (HRA, rent paid, city type, children count, arrears), business 44AD toggle, freelance 44ADA toggle, and F&O gross turnover with live Sec 44AB audit check
taxEngine.js	All tax logic, risk profiler, portfolio generator, projection, tracker savings calc. Pure JS.
api.js	All backend HTTP calls. No API key present anywhere in this file.
Tracker.jsx	Expense tracker: recurring, pagination, CSV/PDF import, budget alerts, edit modal, 8 bank parsers.
Goals.jsx	Goal setting with SIP calculator, instrument recommendation, progress tracking.
Scenarios.jsx	What-if planner with 4 presets, dual chart, milestone table.
ITSections.jsx	20+ IT Act sections guide. Per-section AI advisor calls /ai/tax-advisor.
Login.jsx	Google Sign-in + Email/Password UI with guest skip option.
AuthContext.jsx	Global auth state. Auto-loads Firestore profile on login.
firebase.js	Firestore init + all Firestore CRUD helpers for profiles, transactions, goals, recurring, budgets.
index.css	Kuber/Laxmi theme: saffron gold (#E8921A), midnight maroon (#0e0806), temple emerald (#1DB873).
ITRFiling.jsx	Personalized ITR Filing Guide: auto-determines ITR form, dynamic document checklist, schedule mapper, 8-step portal walkthrough, deadlines/penalty calculator, AI filing assistant. 6 tabs. Props: plannerResults from App.jsx.

7.3 Backend API Routes

Route	Method	Rate Limit	Description
GET /health	GET	None	Server health check
POST /analyze	POST	60/min	Full analysis: multi-income tax + risk + portfolios + projections
POST /ai/explain-portfolio	POST	10/min	Claude explains portfolio with post-tax CAGR and income mix context
POST /ai/tax-advisor	POST	15/min	Free-form tax Q&A. Includes user profile context (income, slab, occupation).

8. Bug Register

All significant bugs encountered during development are documented here for transparency and to demonstrate debugging methodology.

Bug	Root Cause	Fix	Version
Cursor disappears after 1 digit in IncomeForm	SafeInput and SrcGroup defined inside IncomeForm. Every keystroke → parent re-render → React sees new component type → unmounts input → focus lost.	Hoist SafeInput and SrcGroup to module scope (outside function). React reuses stable instances.	v4 fix
Page scrolls to top on keypress	Same as above: input remount caused layout recalculation that shifted scroll position.	Module-scope hoist + 400ms debounce on taxResult useMemo. No layout shift while typing.	v4 fix
Vercel build failure: Unterminated regular expression (Tracker.jsx:433)	Multi-line regex literal in autoCategory() function. JavaScript/esbuild requires all regex patterns to be on a single line.	Collapsed all regex patterns in autoCategory() to single lines. Logic unchanged. Example: /zomato swiggy ... kiit hospitality .../ → one line.	v5 fix
Vercel build failure: React hook called at module scope	export default function Tracker() { ... } wrapper declaration was accidentally deleted during editing. All hooks (useAuth, useState, etc.) were called at module scope — invalid in React.	Re-added export default function Tracker() { declaration before the first hook call. All hooks correctly inside the component function.	v5 fix

8.1 v4 — IncomeForm Cursor & Scroll Root Cause Analysis

Root Cause: Component definitions inside parent functions cause React to treat them as new component types on every render, triggering unnecessary unmounts and remounts.

Both bugs shared the same cause. SafeInput and SrcGroup were defined inside IncomeForm. On every keystroke:

13. The parent IncomeForm re-rendered (state update from typing).
14. New function references were created for SafeInput and SrcGroup.
15. React compared old and new component types — different references = different components.
16. React unmounted the old input and mounted a fresh one — destroying focus.
17. The DOM layout recalculation from the remount caused the page to scroll to the top.

Fix: Both components hoisted to module scope. Their identity is stable across parent re-renders. React reuses the same instances. Additional: taxResult debounced 400ms to prevent recalculation on every keypress.

8.2 v5 — Vercel Build Failures

Both v5 build errors were caught at deployment time by Vercel's esbuild compilation step, not at runtime. Neither was visible in local npm run dev.

Bug 1 — Unterminated Regular Expression: Multi-line regex literals span two or more source lines. The JavaScript/esbuild parser requires all regex patterns to be on a single line — a line break inside /pattern/ is a syntax error. autoCategory() had patterns spanning 2-3 lines for readability during development.

Fix: All regex patterns in autoCategory() collapsed to single lines. Identical logic, no multiline regex literals.

Bug 2 — React Hook at Module Scope: The export default function Tracker() { wrapper declaration was accidentally deleted during a multi-step editing session. All useState, useEffect, useCallback, useMemo, and useRef calls were at module-scope level — valid JavaScript but illegal in React (hooks must be called inside a component or custom hook).

Fix: Re-added export default function Tracker() { before the first hook call. All hooks correctly scoped inside the component function.

9. Deployment

9.1 Infrastructure & Cost

Service	Provider	Tier	Monthly Cost
Frontend	Vercel	Hobby (free)	₹0
Backend	Railway	Hobby (\$5 credit)	₹0
Database + Auth	Firebase (Spark Plan)	Free tier	₹0
AI API	Anthropic	Pay-per-use	~₹0.50 per 100 calls
Total	—	—	₹0–₹67/month

9.2 CI/CD Pipeline

18. Developer edits code and tests locally on localhost:3000 (frontend) and localhost:8000 (backend).
19. Commit and push: git add . && git commit -m "description" && git push
20. Vercel auto-detects the push, rebuilds React app, deploys to CDN (~2-3 minutes).

21. Railway auto-detects the push, rebuilds FastAPI server, restarts (~2-3 minutes).
22. Zero manual steps. Changes are live at the production domain automatically.

10. Comparative Analysis

WealthWise v5 compared against the best feature from each of the 7 referenced competing projects:

Feature	WealthWise v5	Best Competitor
India FY 2026-27 tax engine	Full code implementation	None of 7 projects
New vs Old regime comparison	Yes — with rupee savings + per-regime slab rates	None
Section 87A rebate (Budget 2025)	Implemented	None
HUF entity type	Yes — correct tax treatment	None
Agricultural income (partial integration)	Yes	None
Multi-source income (13 types)	Yes — salary, rental, LTCG, crypto, agri, etc.	None
Loan deductions (home, education)	Yes — old regime	None
Per-asset post-tax CAGR per regime	Yes	None
Dual corpus chart (new vs old)	Yes — 3 lines (pre-tax, new, old)	None
35+ IT Act sections + AI advisor	Complete guide	None
14 Indian + global asset classes	PPF, NPS, ELSS, SGB, Crypto, Savings A/C, etc.	None
Expenditure-based tax deductions	80D from health spending, 80C hints	None
Goal-aware portfolio adjustment	Yes — adjusts allocation by timeline	None
Savings auto-calc from tracker	Yes — toggle manual/auto	None
API key security (backend proxy)	Full — key never in frontend	Mostly exposed in frontend
Firebase Auth (Google + Email)	Yes + guest mode	Mugeshwaran-k (Clerk only)
Expense tracker with AI coach	Yes + family category + deduction panel	Annarhysa/BudgetBud (no AI)
Recurring transactions	Yes — auto-post on due date	None
CSV import from bank statement	Yes — SBI/HDFC/ICICI auto-detect	None

PDF bank statement import	Yes — 8 banks (SBI, KVB, HDFC, ICICI, Axis, Kotak, PNB, BOB)	None
Edit transactions	Yes — inline edit modal	None
Budget alerts per category	Yes — 90% warning, 100% exceeded	None
Goals + SIP + instrument recommendation	Yes + integrated into planner	BudgetBud (basic)
LTCG indexation split (pre/post Jul 2024)	Yes — 20% with indexation vs 12.5% without	None
Scenario planner with milestones	Yes — 4 presets + custom + dual chart	imRahul05 (basic)
PWA / Android installable	Yes (manifest.json)	None
ML model training required	No	EnigmAI (KNN + K-Means)
Kuber/Laxmi visual theme	Yes — saffron, maroon, emerald	None
Personalised ITR filing guide (auto-form, checklist, schedule mapper)	Yes — ITR-1/2/3/4 auto-determined, 8-step portal guide, deadline calculator, AI assistant	None

11. Limitations & Future Work

11.1 Known Limitations

The following limitations are acknowledged and will be addressed in future versions:

- Asset CAGR values are static historical averages — not connected to live AMFI, MCX, or CoinGecko data. All 14 asset class returns are hardcoded research-based estimates.
- LTA (Leave Travel Allowance) is not yet individually input. HRA was resolved in v5.1 via the Sec 10(13A) 3-limit formula with metro/non-metro toggle.
- No Form 16, AIS, or investment statement PDF upload/parsing. Users must manually enter income sources.
- Futures and options are excluded from portfolio allocation as they require specialist knowledge to recommend safely.
- React Native / Expo app for Play Store not yet built; currently PWA-only for Android installation.
- PDF bank statement parsers use regex-based text extraction and may fail on non-standard PDF formatting or password-protected statements.

11.2 Roadmap

Planned Feature	Priority	Effort
LTA (Leave Travel Allowance) inputs	High	1 day
Live MF NAV data via AMFI API	High	1 week
Live FD rates via RBI / bank scraping	High	1 week

Form 16 / AIS PDF upload + parsing (Claude document API)	High	2 weeks
Live stock/MF prices tab (yfinance + AMFI + CoinGecko)	Medium	2 weeks
n8n automation for daily rate updates	Medium	1 week
Crypto tax calculator (30% flat + 1% TDS per transaction)	Medium	3 days
Section 54F and 54B LTCG exemption claim calculator (auto-compute reinvestment amount needed)	Medium	2 days
Open-source LLM option (Groq / Gemini free tier)	Medium	3 days
SIP reminder push notifications (PWA)	Medium	1 week
React Native app via Expo EAS for Play Store	Medium	3 weeks
PyPortfolioOpt (Markowitz MPT) for v6 portfolio	Low	4 weeks
Multi-language support (Hindi, Tamil, Telugu)	Low	3 weeks
DigitalOcean migration + Mumbai CDN	Low	1 week
CSV / PDF export of full financial plan	Medium	1 day

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End of Document — WealthWise Project Documentation v5.0